

Pancreas & Lesion Segmentation

A 3D CADe annotation assist for radiologists — not a diagnostic tool

 pancreas

 lesion

Hand-tracing tumors is slow — and two people are judging

- Radiologists hand-trace the pancreas and any tumor on 3D CT.
- It is slow, tedious work on a small, hard-to-see organ.
- Our tool proposes the outline; the radiologist accepts, edits, or rejects it.

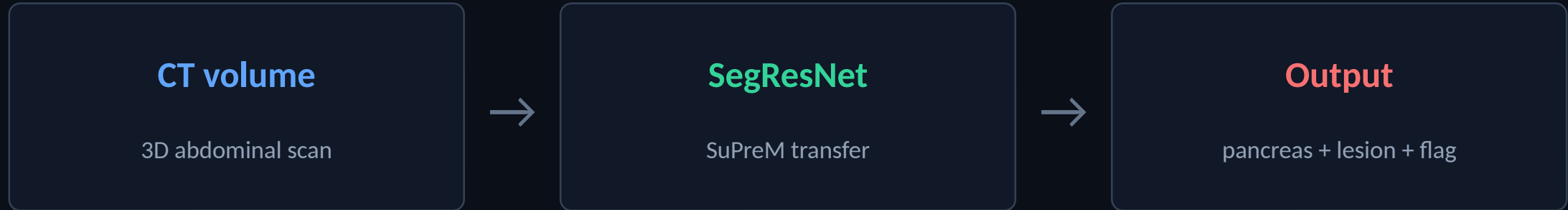
● The technical lead

Are the experiments rigorous, single-variable, and fairly compared?
Is the model choice sound?

● The business stakeholder

What does it do, how accurate is it, and is it trustworthy enough to build on?

A 3D segmentation model with a possible-tumor flag



- Supervised 3D semantic segmentation: background / pancreas / lesion.
- SegResNet (MONAI), fine-tuned from SuPreM pretrained weights — the same Johns Hopkins lab.
- Runs on an Apple-silicon laptop (MPS). Evaluated with full-volume sliding-window inference.

The key decision: feed the whole pancreas box

- Every scan is standardized first — reorient, resample to 1.5 mm, window the intensities, compose the 3-class label. For a vision model, this standardization IS the feature engineering.
- The design decision: instead of random cropped patches, crop to the pancreas and feed the ENTIRE organ as one cube every step — so the model always sees the whole pancreas and its surroundings.

What it bought us

specificity

8% → 55%

lesion Dice

0.21 → 0.26

Two questions, two metrics

Lesion Dice

When there IS a tumor, how well do we outline it?
(accuracy)

Specificity

On a healthy scan, do we correctly stay quiet? (trust)

We score each only where it is meaningful — accuracy on tumor cases, specificity on healthy ones.

The two questions a tumor-finder must answer

Did we catch the cancers (sensitivity), and did we avoid crying wolf (specificity)?



two pull against each other — flag more eagerly and you catch more tumors but raise more false alarms. That trade-off is the operating point we c

First — what didn't work

We tried four recipe changes to raise tumor accuracy. Every one moved it by nothing.

✗	Balanced patch sampling	no change
✗	Loss reweighting	no change
✗	Bigger field of view	no change
✗	Finer resolution	helped the pancreas, not the tumor

Four dead ends — but each one eliminated a suspect.

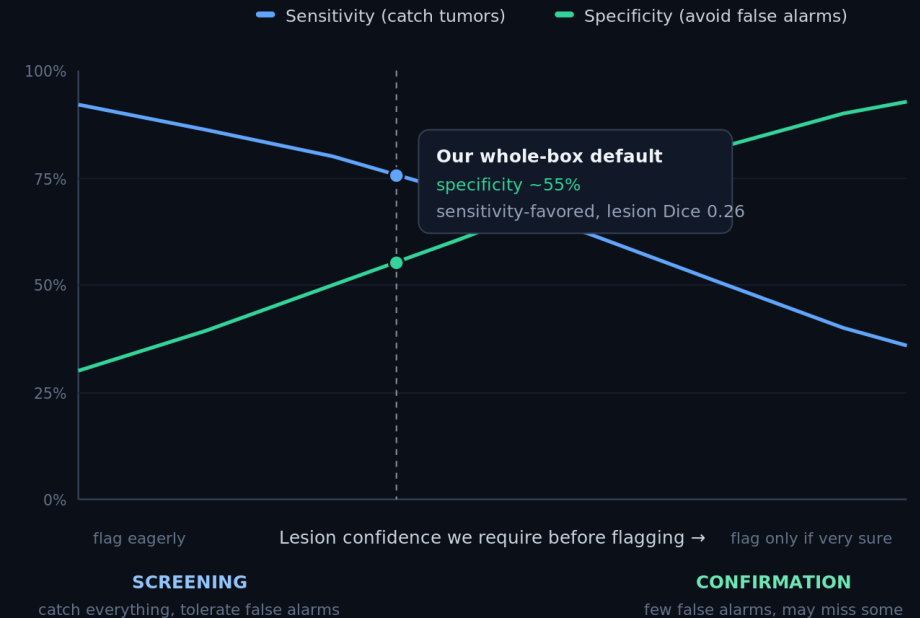
What worked — and the pattern it revealed

- Two things moved the model: a structural change (the whole-box ROI) and pretraining (SuPreM transfer).
- Specificity had many working dials — context, contrast phase, test-time augmentation, threshold.
- But lesion accuracy resisted every recipe knob. All four failures shared one thing: they operated on the same ~50 tumors.

The untested variable was the data itself.

One model, one dial: the operating point we choose

The same model can favor catching tumors or avoiding false alarms — we pick where to sit.



illustrative of the trade-off; the marked point is our measured whole-box result. Levers that move along this dial: threshold, test-time augmentation, c

More data broke the ceiling

lesion Dice

0.26 → 0.33

+0.06 — the first result all week to beat the ceiling four recipe changes couldn't move.

- We held the recipe fixed and tripled the tumor examples (50 → 150). Nothing else changed.
- Confirms it: tumor accuracy was DATA-limited, not recipe-limited.

EXP-17 • data scale-up

	95 cases	300 cases
tumor examples	50	150
lesion Dice	0.263	0.327
pancreas Dice	0.778	0.815
specificity	55%	50%

The model we're taking into Week 4

- Whole-box SuPreM-transfer SegResNet, trained on 300 cases.

pancreas 0.815 • lesion 0.327 • specificity ~50%

Chosen on trade-offs, not just the top metric:

- From scratch scores only 0.12 — so SuPreM transfer is essential, not a marginal boost.
- Honest caveats: an oracle ROI (a provided pancreas box), n=20 evaluation, still data-limited.

Why 0.26 lesion Dice is not "three times worse" than pancreas

Dice is overlap divided by total size, so the same few-voxel miss costs far more on a tiny target.

○ Ground truth ○ Model prediction



ute error, very different score. Read pancreas and lesion Dice separately, and judge the lesion against small-tumor benchmarks, not against th

Ahead of plan, and the path is clear

● Weeks 1-3: done

Data validated, pipeline built, and this week we not only found the ceiling — we broke it.

● Week 4: running tonight

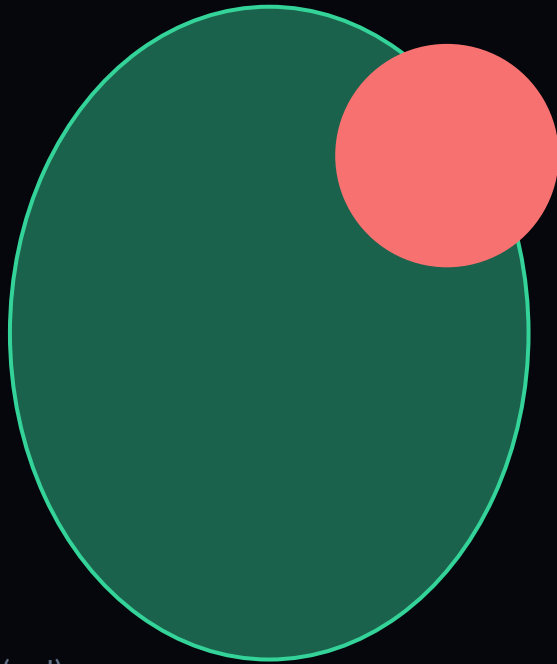
Built a disk cache to scale past 300 cases (600+). Testing whether more data keeps climbing.

● Week 5 + capstone

Failure analysis, the 3D UI, and the autonomous localize-then-segment cascade.

Level 4.5 (pancreas + lesion) is the line that holds. Everything above it is a documented stretch.

What it looks like: an interactive 3D viewer



3D pancreas (green) + lesion (red)

- React + NiiVue, static-first (no backend).
- Rotate freely and cut through the volume in any axis — true 3D, not slice-scrolling.
- Toggle model prediction vs the radiologist's ground truth.
- A plain-language 'possible lesion' summary; the clinical numbers stay in a separate validation view.

Takeaways

- **Nulls are evidence.**

Four dead-end experiments are what made the data conclusion airtight.

- **Structure and transfer make it work; data makes it better.**

Whole-box context and SuPreM pretraining got a usable model; more data broke the accuracy ceiling.

- **On trajectory, honestly framed.**

Best lesion Dice yet (0.33), with clear caveats and a proven Week 4 lever.